

# MONET: Log ratios of Ancestral Variances

A Method for Identifying Local Adaptation in Structured Populations

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## Abstract

Distinguishing adaptive divergence from neutral processes in structured populations is a fundamental challenge in evolutionary biology. Traditional methods like  $Q_{ST}$ - $F_{ST}$  comparisons assume equal relatedness among all subpopulations—an assumption that is often violated in natural metapopulations. When this assumption is broken, these methods can produce inflated false positive rates.

The MONET package implements the LogAV method, which addresses this limitation by explicitly modeling population structure through coancestry matrices. The method compares two estimates of ancestral additive genetic variance derived from different levels of the demographic history. Under neutrality, these estimates should be equal; deviations indicate selection. MONET maintains proper statistical calibration across diverse population structures while achieving comparable or superior power to existing methods.

This vignette provides a practical guide to using MONET, from installation through analysis and interpretation. We demonstrate the workflow with a complete tutorial using simulated data, explain how to prepare your own data, and provide guidance on advanced usage and troubleshooting.

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# 1 Installation and Setup

## 1.1 Installation from GitHub

MONET is available on GitHub. Install it using the `devtools` package:

```
1 # Install devtools if you don't have it
2 if (!require("devtools")) install.packages("devtools")
3
4 # Install MONET from GitHub
5 devtools::install_github("isadoo/MONET")
```

## 1.2 Loading the Package

Once installed, load MONET into your R session:

```
1 library(MONET)
```

## 1.3 Dependencies

MONET automatically installs and uses several key dependencies:

- **brms**: For Bayesian linear mixed-effects modeling. MONET uses **brms** to fit the statistical model and estimate ancestral variances through Bayesian inference.
- **hierfstat**: For calculating coancestry matrices and population genetic statistics using the allele-sharing method of moments.
- **kinship2**: For kinship calculations from pedigree data.
- **dplyr** and **tidyr**: For data manipulation.
- **Matrix**: For efficient matrix operations, particularly when working with large coancestry matrices.
- **posterior**: For posterior distribution analysis from the Bayesian model.

Note that **brms** requires a working Stan installation. If you encounter issues with **brms**, consult the **brms** documentation at <https://paul-buerkner.github.io/brms/>.

# 2 Quick Overview

## 2.1 The MONET Workflow

MONET follows a three-step workflow:

1. **Calculate coancestries**: Use the `calculate_coancestries()` function to compute population-level ( $\Theta_P$ ) and individual-level ( $\mathbf{M}$ ) coancestry matrices from genetic data.
2. **Run the analysis**: Use the `monet()` function to fit a Bayesian mixed-effects model to your trait data, accounting for population structure.
3. **Interpret results**: Examine the log-ratio of ancestral variances and its associated p-value to assess evidence for local or global adaptation.

## 2.2 The Core Statistical Test

MONET tests for local adaptation by comparing two estimates of the ancestral additive genetic variance ( $V_A$ ):

- $\hat{V}_{A,B}$ : estimated from between-population divergence
- $\hat{V}_{A,W}$ : estimated from within-population variation

Under neutrality, these two estimates should be equal on average:  $E[V_{A,B}] = E[V_{A,W}]$ , as both reflect the same ancestral variance partitioned through different aspects of the demographic history.

The test statistic is the log-ratio:

$$\text{LogAV} = \log \left( \frac{\hat{V}_{A,B}}{\hat{V}_{A,W}} \right)$$

Under neutrality,  $E[\text{LogAV}] = 0$ .

- If  $\text{LogAV}$  is significantly  $> 0$  (i.e.,  $V_{A,B} > V_{A,W}$ ): This indicates **local adaptation**. Phenotypic divergence among populations exceeds neutral expectations.
- If  $\text{LogAV}$  is significantly  $< 0$  (i.e.,  $V_{A,B} < V_{A,W}$ ): This indicates **global (or spatially-uniform) adaptation**. Selection has reduced divergence among populations relative to what would be expected under drift alone.

## 2.3 Statistical Inference

MONET uses Bayesian inference to compute the log-ratio. From the posterior distributions of  $V_{A,B}$  and  $V_{A,W}$ , we calculate the log-ratio for each posterior sample. The Bayesian p-value is computed as:

$$p = 2 \times \min \left( \frac{\#(\text{LogAV} > \text{median})}{N}, \frac{\#(\text{LogAV} < \text{median})}{N} \right)$$

where  $N$  is the number of posterior samples, this p-value quantifies the proportion of the posterior in which the log-ratio has the opposite sign to the median, effectively testing whether the log-ratio differs significantly from zero.

A small p-value (typically  $p < 0.05$ ) provides evidence against neutrality and supports the presence of selection.

## 2.4 Testing Environmental Correlations

MONET can also test whether trait values are linearly associated with environmental differences across populations. For that, environmental covariates should be included in the model (via the `formula_covariates` parameter). This test uses the coefficient  $\beta$  from the fixed effects in the model.

For each environmental variable  $j$ , MONET computes a Bayesian p-value:

$$p_j = 2 \times \min \left( \frac{\#(\beta_j > 0)}{N}, \frac{\#(\beta_j < 0)}{N} \right) \quad (1)$$

This tests whether the environmental coefficient differs significantly from zero. A significant coefficient indicates that trait values systematically vary with that environmental variable across populations, even after accounting for neutral population structure through the random effects.

Both tests (LogAV and the environmental coefficient) can be significant simultaneously, providing both evidence for selection (LogAV) and insight into the ecological driver (environmental coefficient).

## 3 Data Preparation

### 3.1 Data Requirements

To use MONET, you need:

1. **Phenotypic data from a common garden:** Quantitative trait measurements from offspring (F1 generation) reared in a controlled environment to minimize phenotypic plasticity.
2. **Genetic markers from the parental (founder) generation:** Neutral markers (SNPs, microsatellites, etc.) from individuals sampled from wild populations. This is used to estimate the population-level coancestry matrix  $\Theta_P$ .
3. **Relatedness information for F1 individuals:** Either:
  - Genetic data from F1 individuals (for estimating kinship), or
  - Pedigree information (dam and sire IDs for each F1 individual)
4. **Population identification:** A mapping that links both parental and F1 individuals to their respective source populations.
5. **(Optional) Environmental data:** Environmental covariates measured at each source population can be included to test hypotheses about specific ecological drivers of divergence and to increase statistical power.

### 3.2 Data Format Requirements

#### 3.2.1 Genetic Data (Dosage Format)

Genetic marker data must be in **dosage format**, where each genotype is encoded as:

- **0:** homozygous for the reference allele
- **1:** heterozygous
- **2:** homozygous for the alternative allele

The data should be organized as a matrix or data frame with:

- **Rows:** individuals
- **Columns:** loci (genetic markers)

For example, if you have  $n$  individuals genotyped at  $L$  loci, your dosage matrix should have dimensions  $n \times L$ .

**Example structure:**

```

1 head(genetic_data_parents[, 1:6])
2 #   locus1 locus2 locus3 locus4 locus5
3 #     1      0      2      1      0
4 #     0      1      1      2      1
5 #     2      2      0      0      2

```

Row names should be individual IDs, and column names should be locus IDs.

### 3.2.2 Population Identification

Population information for the parental generation should be provided as a **vector** of population IDs, with one entry per individual, in the same order as the rows in the genetic data:

```

1 # Example for 6 individuals from 3 populations
2 parent_populations <- c("pop1", "pop1", "pop2", "pop2", "pop3", "pop3")

```

For the F1 generation, provide a **data frame** with at least two columns:

- One column for individual IDs
- One column for population IDs

```

1 head(population_individual_id)
2 #   id      population_id
3 #   F001      pop1
4 #   F002      pop1
5 #   F003      pop1
6 #   F004      pop2

```

### 3.2.3 Phenotypic Data

Trait data should be a data frame with at least two columns:

- One column for individual IDs (matching the F1 individuals)
- One column for trait values

```

1 head(trait_data)
2 #   id      trait_value
3 #   F001      15.2
4 #   F002      14.8
5 #   F003      16.1
6 #   F004      13.9

```

You can include additional columns for:

- Population IDs (if not provided separately)
- Environmental covariates
- Other fixed effects (e.g., sex, age, block effects)
- Standard errors for trait measurements (if available)

### 3.2.4 Pedigree Data (if using instead of F1 genetic data)

If you are using pedigree information instead of F1 genetic data, provide a data frame with at least these columns:

- Individual ID
- Dam (mother) ID
- Sire (father) ID
- Dam population ID
- Sire population ID

## 3.3 Converting VCF to Dosage Format

If your genetic data is in VCF (Variant Call Format), you need to convert it to dosage format. One way is through **hierfstat** `read.VCF` function:

```
1 bed <- hierfstat::read.VCF
2 dos <- gaston::as.matrix(bed)
```

Some of the many other approaches include: (1) `vcftools` Danecek et al. [2011] and R. (2) the package `vcfR` in R Knaus and Grünwald [2017]. (3) `PLINK` Weeks [2010] and R.

## 3.4 Reconstructing Parental Genotypes

If you have genetic data only from the F1 generation and lack parental genetic data, you may be able to reconstruct parental genotypes using specialized software. This is particularly useful when you have pedigree information but did not genotype the parental generation. Some of these software include `COLONY` Jones and Wang [2010], and `Sequoia` Huisman [2017].

**Important considerations:** Here are some considerations when using parental reconstructions, but we recommend reading the author’s recommendations for each software.

- Parental genotype reconstruction is most reliable when you have:
  - High marker density (many SNPs)
  - Multiple offspring per parent
  - Known pedigree structure
- Reconstruction quality varies by individual—some parents may be confidently reconstructed while others remain uncertain
- When using reconstructed genotypes, be aware that reconstruction errors can affect coancestry estimates.

## 4 Environmental information

Including environmental covariates in your analysis can substantially increase statistical power to detect selection while simultaneously providing insight into the ecological drivers of divergence.

Traditional approaches to studying local adaptation follow a two-step process: First testing for patterns of non-neutral divergence (for example through  $Q_{ST} - F_{ST}$ ), and then through a separate analysis, regress population trait means against environmental variables. However, this approach does not leverage environmental information in the test for divergence, and the regression does not account for the non-independence between subpopulations.

MONET uses a mixed-effect framework which allows to address both steps simultaneously. It does so by incorporating environmental covariates directly into the model which is accounting for the population structure. In our benchmarking we observed substantial power gains when including environmental information (See paper [\[how do we refer to it?\]](#)). Importantly, these power gains do not inflate false positive rates.

When environmental covariates are included, MONET provides two complementary pieces of information: LogAV test which describes potential evidence for selection and the environmental coefficients which describes the direction and strength of the association between trait and environmental variation.

## 5 Tutorial with Example Data

### 5.1 Overview of the Example Dataset

This tutorial uses a simulated dataset designed to demonstrate MONET's workflow.

- **Population structure:** 18 populations arranged in a stepping-stone configuration, where gene flow only occurs between adjacent populations
- **Sample sizes:**
  - Parental generation: 10 individuals per population (180 total)
  - F1 generation: 50 individuals per population (900 total)
- **Selection scenario:** Local adaptation following a parabola-shaped environmental distribution of the population. In Figure 1, you can see the trait distribution, in which the traits well represent the optima the subpopulations were exposed to as shown in Figure 2.
- **Genetic markers:** 2,000 neutral SNPs used for coancestry estimation
- **Crossing design:** Within-population crosses producing F1 offspring

### 5.2 Loading Tutorial Data

The tutorial data is included with the MONET package and can be loaded directly from the package installation. We will load genetic data for both the parental and F1 generations, trait measurements, and population identifiers.

```
1 # Step 1: Load genetic data (dosage format)
2 # Load parental dosages
3 dos_P_neutral <- readRDS(
4   system.file("extdata", "vignette_dos_parental_neutral.rds",
```

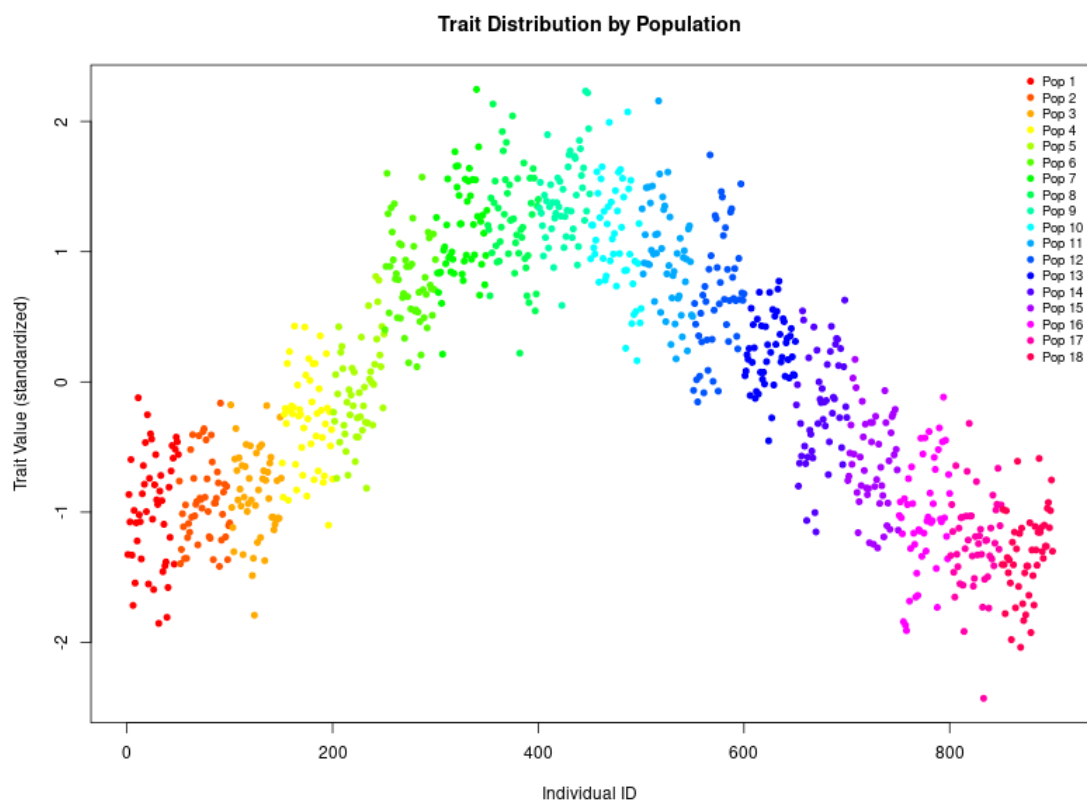


Figure 1: Trait value by individual with population identification in different colours.

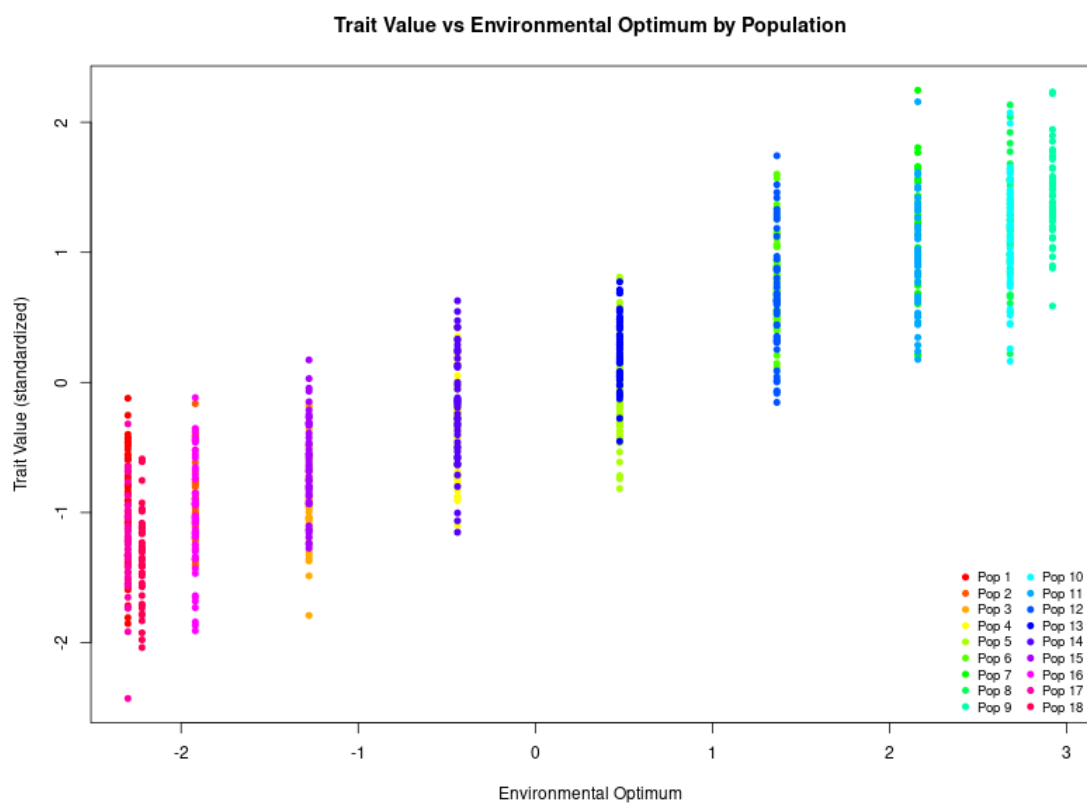


Figure 2: Comparison between individual trait values and optima with subpopulations differentiated by colour.

```

5         package = "MONET")
6     )
7
8     # Load F1 dosages
9     dos_F1only_neutral <- readRDS(
10         system.file("extdata", "vignette_dos_F1only_neutral.rds",
11                     package = "MONET")
12     )
13
14     # Check the format
15     cat("Parental dosage format - first 5 individuals, first 5 loci:\n")
16     print(dos_P_neutral[1:5, 1:5])

```

### Output:

Parental dosage format - first 5 individuals, first 5 loci:

|      | n1_l1 | n1_l2 | n1_l3 | n1_l4 | n1_l5 |
|------|-------|-------|-------|-------|-------|
| [1,] | 2     | 0     | 1     | 1     | 1     |
| [2,] | 2     | 0     | 1     | 1     | 1     |
| [3,] | 2     | 0     | 1     | 1     | 1     |
| [4,] | 2     | 0     | 2     | 1     | 1     |
| [5,] | 2     | 0     | 2     | 1     | 1     |

```

1 # F1 dosage data
2 cat("F1 dosage format - first 5 individuals, first 5 loci:\n")
3 print(dos_F1only_neutral[1:5, 1:5])
4 cat("\nF1 neutral dosages dimensions:", dim(dos_F1only_neutral), "\n")

```

### Output:

F1 dosage format - first 5 individuals, first 5 loci:

|      | [,1] | [,2] | [,3] | [,4] | [,5] |
|------|------|------|------|------|------|
| [1,] | 2    | 0    | 2    | 1    | 0    |
| [2,] | 2    | 0    | 2    | 1    | 0    |
| [3,] | 2    | 0    | 2    | 0    | 0    |
| [4,] | 2    | 0    | 1    | 2    | 1    |
| [5,] | 2    | 0    | 2    | 1    | 0    |

F1 neutral dosages dimensions: 900 2000

The genetic data is in the correct dosage format (0, 1, 2) with individuals as rows and loci as columns. Next, we load the phenotypic trait data and population identifiers:

```

1 # Step 2: Load trait data
2 trait_df_pop <- read.csv(
3     system.file("extdata", "vignette_trait_df_pop.csv",
4                 package = "MONET")
5 )
6
7 # Load population-to-individual mapping (F1 individuals)
8 population_individual_id <- read.csv(
9     system.file("extdata", "vignette_population_individual_id_df.csv",
10                 package = "MONET")

```

```

11 )
12
13 # Load population IDs for genotyped parents
14 pop_id <- readRDS(
15   system.file("extdata", "vignette_pop_id.rds",
16               package = "MONET")
17 )
18
19 # Examine the trait data
20 cat("Trait data (first 6 rows):\n")
21 print(head(trait_df_pop))
22
23 cat("\nPopulation-individual mapping (first 6 rows):\n")
24 print(head(population_individual_id))

```

### Output:

Trait data (first 6 rows):

|   | individual | trait      | population |
|---|------------|------------|------------|
| 1 | 1          | -1.3250641 | 1          |
| 2 | 2          | -0.8642926 | 1          |
| 3 | 3          | -1.0746377 | 1          |
| 4 | 4          | -0.5960614 | 1          |
| 5 | 5          | -1.3298046 | 1          |
| 6 | 6          | -1.7154481 | 1          |

Population-individual mapping (first 6 rows):

|   | pop_id | individual |
|---|--------|------------|
| 1 | 1      | 1          |
| 2 | 1      | 2          |
| 3 | 1      | 3          |
| 4 | 1      | 4          |
| 5 | 1      | 5          |
| 6 | 1      | 6          |

The trait data includes individual IDs, trait values, and population assignments.

## 5.3 Calculating Coancestries

Now we calculate the two coancestry matrices needed for MONET:  $\Theta_P$  (population-level) and  $M$  (individual-level). The `calculate_coancestries()` function handles all the computation:

```

1 # Step 3: Calculate coancestry matrices
2 coancestries_dosage <- calculate_coancestries(
3   genetic_data_parents      = dos_P_neutral,
4   genotyped_parent_populations = pop_id,
5   genetic_data_F1          = dos_F1only_neutral,
6   population_individual_id   = population_individual_id,
7   column_individual         = "individual",
8   column_population         = "pop_id",
9   all_parents_genotyped     = TRUE
10 )

```

### Output:

```
There are 18 populations.
Calculating Theta.P
Theta.P calculated with dimensions 18 18
Calculating M
population sizes of F1 50 50 50 50 50 50 50 50 50 50 50 50 50 50 50 50 50
total individuals are 900
pop: 0
pop: 1
pop: 2
...
pop: 17
There are 50 individuals in population 1 for calculating the M matrix
There are 50 individuals in population 2 for calculating the M matrix
...
There are 50 individuals in population 18 for calculating the M matrix
M calculated with dimensions 900 900
```

Now let's extract them and check their properties:

```
1 # Extract the matrices
2 Theta.P <- coancestries_dosage$Theta.P
3 M <- coancestries_dosage$M
4
5 cat("\nCoancestry matrices calculated:\n")
6 cat("  Theta.P dimensions:", dim(Theta.P), "\n")
7 cat("  M dimensions:", dim(M), "\n")
```

### Output:

```
Coancestry matrices calculated:
  Theta.P dimensions: 18 18
  M dimensions: 900 900
```

If either matrix is not positive definite (minimum eigenvalue  $< 0$ ), MONET will automatically correct it using `nearPD` and issue a warning. However, it is important to note that non-positive-definiteness can signal issues in the estimation of relatedness. If you receive a non-positive-definite warning, ensure that your individuals are correctly attributed to their populations of origin and double-check individual identification. If there are very negative eigenvalues, the correction from `nearPD` might require large distortions of the relatedness patterns.

## 5.4 Running MONET

### 5.4.1 Basic Analysis Without Covariates

Now we run the main MONET analysis. This fits a Bayesian mixed-effects model to test whether the trait shows evidence of local adaptation:

```
1 # Step 4: Run MONET analysis
2 results <- monet(
3   Theta.P = Theta.P,
4   M = M,
```

```

5 trait_dataframe = trait_df_pop,
6 column_individual = "individual",
7 column_trait = "trait",
8 iter = 4000,
9 control = list(adapt_delta = 0.95, max_treedepth = 12)
10 )

```

The function will print information about the model being fit:

#### Initial output:

Using formula:  $Y \sim 1 + (1 \mid \text{gr}(\text{pop}, \text{cov} = \text{two.Theta.P})) + (1 \mid \text{gr}(\text{ind}, \text{cov} = \text{M}))$

Total rows: 900 ; complete rows used: 900

Compiling Stan program...

Start sampling

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 1).

Chain 1:

Chain 1: Gradient evaluation took 0.000544 seconds

Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 5.44 seconds.

Chain 1: Adjust your expectations accordingly!

...

The model will take a few minutes to run. After sampling completes, MONET prints a model summary:

#### Model summary output:

=== Model Summary ===

Family: gaussian

Links: mu = identity

Formula:  $Y \sim 1 + (1 \mid \text{gr}(\text{pop}, \text{cov} = \text{two.Theta.P})) + (1 \mid \text{gr}(\text{ind}, \text{cov} = \text{M}))$

Data: dat (Number of observations: 900)

Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 2;

total post-warmup draws = 4000

Multilevel Hyperparameters:

~ind (Number of levels: 900)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|------|----------|----------|
| sd(Intercept) | 0.39     | 0.03      | 0.34     | 0.45     | 1.01 | 278      | 468      |

~pop (Number of levels: 18)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|------|----------|----------|
| sd(Intercept) | 1.13     | 0.23      | 0.79     | 1.68     | 1.00 | 2384     | 3118     |

Regression Coefficients:

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| Intercept | -0.30    | 0.66      | -1.60    | 1.02     | 1.00 | 2947     | 3212     |

Further Distributional Parameters:

|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 0.22     | 0.02      | 0.17     | 0.26     | 1.02 | 243      | 315      |

### Understanding the model output:

- **sd(Intercept) for  $\sim\text{pop}$ :** This is the estimated standard deviation of the population random effect, which equals  $\sqrt{V_{A,B}}$ . Here: 1.13, so  $V_{A,B} \approx 1.13^2 = 1.28$ .
- **sd(Intercept) for  $\sim\text{ind}$ :** This is the estimated standard deviation of the individual random effect, which equals  $\sqrt{V_{A,W}}$ . Here: 0.39, so  $V_{A,W} \approx 0.39^2 = 0.15$ .
- **sigma:** Residual standard deviation (environmental/non-additive genetic variation)

### 5.4.2 Interpreting the Results

Now let's examine the MONET-specific output:

```
1 # View the results
2 print(results)
```

#### Output:

```
=== MONET Analysis Results ===
```

```
Trait analyzed: trait
```

```
Formula used: Y ~ 1 + (1 | gr(pop, cov = two.Theta.P)) + (1 | gr(ind, cov = M))
```

```
--- Log-Ratio of Ancestral Variances ---
```

```
Mean: 2.0713
```

```
Median: 2.0514
```

```
95% Credible Interval: [1.3159, 2.9515]
```

```
P-value: 0.0000
```

```
--- Convergence Diagnostics ---
```

```
Max R-hat: 1.0167 (WARNING: > 1.01)
```

```
Divergent transitions: 0
```

Use `plot()` to visualize the posterior distribution.

#### Key results explained:

- **Log-Ratio Mean/Median (2.07/2.05):** The log-ratio  $\log(V_{A,B}/V_{A,W})$  is substantially greater than zero. This is computed from the posterior samples: for each MCMC sample, we calculate  $\log(\text{sd\_pop}^2/\text{sd\_ind}^2)$ .
- **95% Credible Interval [1.32, 2.95]:** There is 95% probability that the log-ratio lies in this range. Importantly, this interval does not include zero, providing strong evidence against neutrality.
- **P-value: 0:** This is the Bayesian p-value testing  $H_0: \text{LogAV} = 0$ . A value this small provides evidence that it is unlikely drift alone would have caused the observed patterns of divergence. Note that the lowest values depend on the total number of iterations. Here, we can say that  $p \leq 1/4000 = 0.00025$

- **Interpretation:** Since  $\text{LogAV}$  is significantly  $> 0$  (i.e.,  $V_{A,B} > V_{A,W}$ ), it indicates **local adaptation**. This result, together with the p-value, points to populations having diverged in this trait more than expected under neutral evolution alone, suggesting that selection has driven phenotypic differentiation.

### Examining posterior samples:

You can inspect the raw posterior samples that MONET saved:

```
1 # First few posterior samples
2 print(head(results$sampling))
```

### Output:

```
# A tibble: 6 x 6
  b_Intercept sd_pop__Intercept sd_ind__Intercept V_AB V_AW log_ratio
      <dbl>          <dbl>          <dbl> <dbl> <dbl> <dbl>
1    -0.965          1.54          0.409 2.38  0.167  2.65
2    -0.607          0.998          0.416 0.995 0.173  1.75
3    -1.31           1.25          0.426 1.57  0.181  2.16
4     0.384          1.25          0.425 1.57  0.181  2.16
5    -0.103          1.11          0.427 1.23  0.182  1.91
6    -0.0256         0.720          0.432 0.518 0.187  1.02
```

Each row is one MCMC sample. The columns are:

- **b\_Intercept:** Overall mean (varies due to sampling uncertainty)
- **sd\_pop\_\_Intercept:**  $\sqrt{V_{A,B}}$  for this sample
- **sd\_ind\_\_Intercept:**  $\sqrt{V_{A,W}}$  for this sample
- **V\_AB:**  $V_{A,B}$  (additive ancestral variance got from the between-subpopulation divergence)
- **V\_AW:**  $V_{A,W}$  (additive ancestral variance got from the variation within subpopulations)
- **log\_ratio:**  $\log(V_{A,B}/V_{A,W})$  for this sample

### Visualizing the posterior:

```
1 # Plot the posterior distribution of the log-ratio
2 plot(results)
```

This produces a density plot of the log-ratio posterior distribution (Figure 3 , showing:

- The median (dashed line)
- The 95% credible interval (dotted lines)
- The full posterior density

Because the distribution is well above zero, we have strong evidence for local adaptation (See Figure 3).

### Additional components in the results object:

The results object contains:

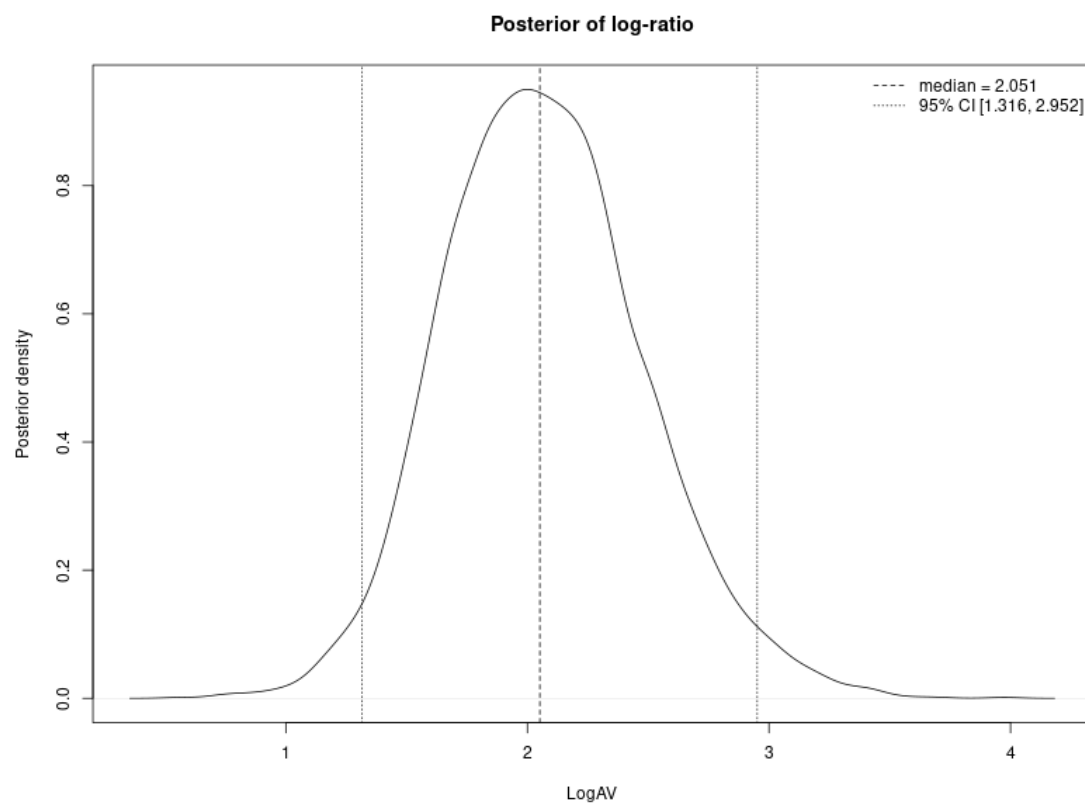


Figure 3: Posterior distribution of LogAV.

- `sampling`: Posterior samples (data frame if `save_full_model=FALSE`, `brmsfit` object if `TRUE`)
- `log_ratio`: Summary statistics for the log-ratio
- `hypothesis`: Results from `brms` hypothesis testing
- `trait_name`: Name of the analyzed trait
- `formula_used`: The model formula
- `convergence`: Convergence diagnostics (Rhat, divergences)

## 5.5 Analyzing Environmental Covariate

```
1 # Step 5: Load environmental data and run MONET with environment
2 trait_df_pop_env <- read.csv(
3   system.file("extdata", "vignette_environment_df.csv", package = "MONET")
4 )
```

```
1 print(head(trait_df_pop_env))
```

Output:

|   | individual | trait      | population | environment |
|---|------------|------------|------------|-------------|
| 1 | 1          | -1.3250641 | 1          | -5          |
| 2 | 2          | -0.8642926 | 1          | -5          |
| 3 | 3          | -1.0746377 | 1          | -5          |
| 4 | 4          | -0.5960614 | 1          | -5          |
| 5 | 5          | -1.3298046 | 1          | -5          |
| 6 | 6          | -1.7154481 | 1          | -5          |

```
1 #Running MONET analysis with environmental covariate
2 results_env <- monet(
3   Theta.P = Theta.P,
4   M = M,
5   trait_dataframe = trait_df_pop_env,
6   column_individual = "individual",
7   column_trait = "trait",
8   formula_covariates = "environment",
9   save_full_model = FALSE,
10  iter = 4000,
11  warmup = 2000,
12  control = list(adapt_delta = 0.95, max_treedepth = 12)
13 )
```

Output:

Multilevel Hyperparameters:

~ind (Number of levels: 900)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|------|----------|----------|
| sd(Intercept) | 0.40     | 0.03      | 0.35     | 0.47     | 1.01 | 206      | 246      |

~pop (Number of levels: 18)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|------|----------|----------|
| sd(Intercept) | 0.26     | 0.15      | 0.02     | 0.58     | 1.01 | 887      | 1650     |

Regression Coefficients:

|             | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept   | -0.07    | 0.18      | -0.45    | 0.29     | 1.00 | 2738     | 2332     |
| environment | 0.48     | 0.04      | 0.38     | 0.56     | 1.00 | 3200     | 2373     |

Further Distributional Parameters:

|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 0.21     | 0.02      | 0.16     | 0.25     | 1.01 | 197      | 214      |

## 6 Advanced Usage

### 6.1 Including Additional Covariates

Beyond environmental variables, you may want to include other covariates in your model to account for confounding factors or experimental design elements. These could be biological ones (sex, age, body size) or experimental design factors (common garden block, measurement year, rearing container).

#### How to include covariates:

Covariates should be columns in your trait data frame, which should be the same name you include in your covariates formula. You specify them using the `formula_covariates` parameter:

```

1 # Example: Including sex as a covariate
2 result_with_sex <- monet(
3   Theta.P      = Theta.P,
4   M            = M,
5   trait_dataframe = trait_data,
6   column_individual = "id",
7   column_trait    = "trait_value",
8   column_population = "population",
9   formula_covariates = "sex"
10 )
11
12 # Multiple covariates
13 result_multiple <- monet(
14   Theta.P      = Theta.P,
15   M            = M,
16   trait_dataframe = trait_data,
17   column_individual = "id",
18   column_trait    = "trait_value",
19   formula_covariates = "sex + age + block"
20 )

```

#### Important notes:

- Continuous covariates are automatically standardized (mean-centered and scaled by standard deviation) before fitting
- Categorical covariates should be coded as factors in R

- The model estimates fixed effects for each covariate while still accounting for population structure through the random effects
- You can examine covariate effects in the model summary

### 6.1.1 Custom Model Specifications

MONET uses `brms` for model fitting, which provides extensive flexibility for customization. You can pass additional arguments to `brms` through the `monet()` function.

#### Custom Priors:

By default, MONET uses the default priors from `brms`. However, you may want to specify custom priors based on your system or prior knowledge, doing the following:

```

1 library(brms)
2
3 # Define custom priors
4 my_priors <- c(
5   prior(normal(0, 1), class = sd, group = pop),
6   prior(normal(0, 1), class = sd, group = ind),
7   prior(normal(0, 0.5), class = sigma)
8 )
9
10 # Run MONET with custom priors
11 result_custom <- monet(
12   Theta.P      = Theta.P,
13   M             = M,
14   trait_dataframe = trait_data,
15   column_individual = "id",
16   column_trait   = "trait_value",
17   prior         = my_priors # passed to brms
18 )

```

#### MCMC Settings:

You can adjust the MCMC sampling parameters for better convergence or faster computation:

```

1 # More iterations for difficult models
2 result_long <- monet(
3   Theta.P      = Theta.P,
4   M             = M,
5   trait_dataframe = trait_data,
6   column_individual = "id",
7   column_trait   = "trait_value",
8   iter          = 8000,      # default: 5000
9   warmup        = 3000,     # default: 2000
10  thin           = 3,        # default: 2
11  chains         = 6
12 )
13
14 # Adjust adapt_delta for divergent transitions
15 result_adapted <- monet(
16   Theta.P = Theta.P,
17   M = M,
18   trait_dataframe = trait_data,
19   column_individual = "id",

```

```

20 column_trait = "trait_value",
21 control = list(adapt_delta = 0.99) # default: 0.8
22 )

```

### Measurement Error:

If you have standard errors for your trait measurements, include them to account for measurement uncertainty:

```

1 # Your trait data should have an SE column
2 head(trait_data)
3 #   id      trait_value  trait_se
4 #   F001      15.2        0.5
5 #   F002      14.8        0.4
6
7 result_with_se <- monet(
8   Theta.P      = Theta.P,
9   M            = M,
10  trait_dataframe = trait_data,
11  column_individual = "id",
12  column_trait    = "trait_value",
13  column_se       = "trait_se" # specify SE column
14 )

```

### 6.1.2 Model Fit

Bayesian models require careful checking of convergence diagnostics to ensure the results are reliable.

#### Checking the MONET output:

The MONET results object includes basic convergence diagnostics:

```

1 # After running monet()
2 result$convergence
3
4 # $n_divergent
5 # [1] 0
6 #
7 # $max_rhat
8 # [1] 1.002

```

#### Accessing the full brms model:

For further analysis and diagnostics, save the full brms model:

```

1 result_full <- monet(
2   Theta.P      = Theta.P,
3   M            = M,
4   trait_dataframe = trait_data,
5   column_individual = "id",
6   column_trait    = "trait_value",
7   save_full_model = TRUE
8 )

```

## 7 Theoretical Background

### 7.1 The Problem with $Q_{ST}$ - $F_{ST}$

The classical approach to detecting local adaptation in quantitative traits is the  $Q_{ST}$ - $F_{ST}$  comparison [Spitze, 1993]. This method compares quantitative trait differentiation, measured through  $Q_{ST}$ , to the neutral expectation calculated from neutral markers ( $F_{ST}$ ). Under neutrality,  $Q_{ST}$  should equal  $F_{ST}$  on average.

$Q_{ST}$  is defined as:

$$Q_{ST} = \frac{V_B}{V_B + 2V_W}$$

where  $V_B$  is the additive genetic variance between populations and  $V_W$  is the additive genetic variance within populations.

The issue we were focusing on addressing with monet lies on a critical assumption of  $Q_{ST}$ - $F_{ST}$  methods.  $Q_{ST}$ - $F_{ST}$  methods assume that all subpopulations are equally related, that is, they follow an island model with uniform migration rates and equal effective population sizes. Under this assumption,  $F_{ST}$  is a scalar that adequately describes population structure.

However, metapopulations are not likely to follow an island model. Instead, we expect metapopulations to have isolation by distance, potentially hierarchical structures with some subpopulations clustering into distinct lineages, variable effective sizes with some subpopulations having higher inbreeding than others, and so on.

When population structure deviates from the island model,  $Q_{ST}$ - $F_{ST}$  comparisons can produce inflated false positive rates [de Villemereuil et al., 2022, do O et al., 2025]. This occurs because the method incorrectly assumes equal relatedness between all population pairs when calculating the expected neutral divergence.

MONET addresses the limitations of  $Q_{ST}$ - $F_{ST}$  by explicitly modeling population structure using coancestry matrices rather than relying on a single summary statistic.

The method is based on the premise that, under neutrality, we can estimate the ancestral additive genetic variance in two independent ways, and these estimates should agree.

Consider a metapopulation that descended from a common ancestral population. The ancestral population had some additive genetic variance  $V_A$ . As populations diverged through drift and limited gene flow, this ancestral variance became partitioned into between-population and within-population components.

We can estimate  $V_A$  using:

1. **Between-population information** ( $\hat{V}_{A,B}$ ): How populations diverged from each other
2. **Within-population information** ( $\hat{V}_{A,W}$ ): How individuals within populations are related

Under neutrality:  $E[V_{A,B}] = E[V_{A,W}] = V_A$

Under selection:

- **Local adaptation:**  $V_{A,B} > V_{A,W}$  (populations diverged more than expected)
- **Global adaptation:**  $V_{A,B} < V_{A,W}$  (populations converged due to similar selection)

#### 7.1.1 The Statistical Model

MONET fits a Bayesian linear mixed-effects model (through `brms` to the phenotypic data:

$$z = \mu + X\beta + a_p + a_i + \epsilon$$

where:

- $z$  is the vector of phenotypic measurements (standardized)
- $\mu$  is the overall mean
- $X\beta$  represents fixed effects (environmental covariates, experimental factors, etc.)
- $a_p$  is the population-level random effect
- $a_i$  is the individual-level random effect
- $\epsilon$  is residual error

The random effects are distributed as:

$$a_p \sim N(0, 2\Theta_p V_{\mathcal{A},B}) \quad (2)$$

$$a_i \sim N(0, M V_{\mathcal{A},W}) \quad (3)$$

The model simultaneously estimates  $V_{\mathcal{A},B}$  and  $V_{\mathcal{A},W}$ , along with all other parameters, using Markov Chain Monte Carlo (MCMC) sampling.

### 7.1.2 The Test Statistic

The log-ratio of the two variance estimates provides our test statistic:

$$\text{LogAV} = \log \left( \frac{\hat{V}_{\mathcal{A},B}}{\hat{V}_{\mathcal{A},W}} \right)$$

The Bayesian framework provides a full posterior distribution for LogAV, from which we compute the p-value.

## 7.2 Coancestry Matrices

MONET uses two coancestry matrices to model population structure:

### 7.2.1 Population-Level Coancestry ( $\Theta_P$ )

$\Theta_P$  is an  $r \times r$  matrix (where  $r$  is the number of populations) that describes relatedness among populations.

Element  $\Theta_{P,ij}$  represents the average probability of identity by descent (IBD) between a random allele from population  $i$  and a random allele from population  $j$ , relative to the ancestral population.

**Calculation:** MONET uses the allele-sharing method of moments [Weir and Goudet, 2017, Goudet and Weir, 2023]:

$$\hat{\Theta}_P = \frac{\bar{A}_s^{\mathcal{P}} - \min(\bar{A}_s^{\mathcal{P}})}{1 - \min(\bar{A}_s^{\mathcal{P}})}$$

where  $\bar{A}_s^{\mathcal{P}}$  is a matrix of population-level allele-sharing statistics. Diagonal elements  $\Theta_{P,ii}$  represent average coancestry within population  $i$ . Off-diagonal elements  $\Theta_{P,ij}$  represent coancestry between populations  $i$  and  $j$ . Higher values indicate closer relatedness (more recent common ancestry or more gene flow). The matrix captures the complete demographic history and structure.

In a stepping-stone model, like the one we used in our tutorial section, you would expect populations that are closer together to have higher coancestry values, which is what we show in Figure 4.

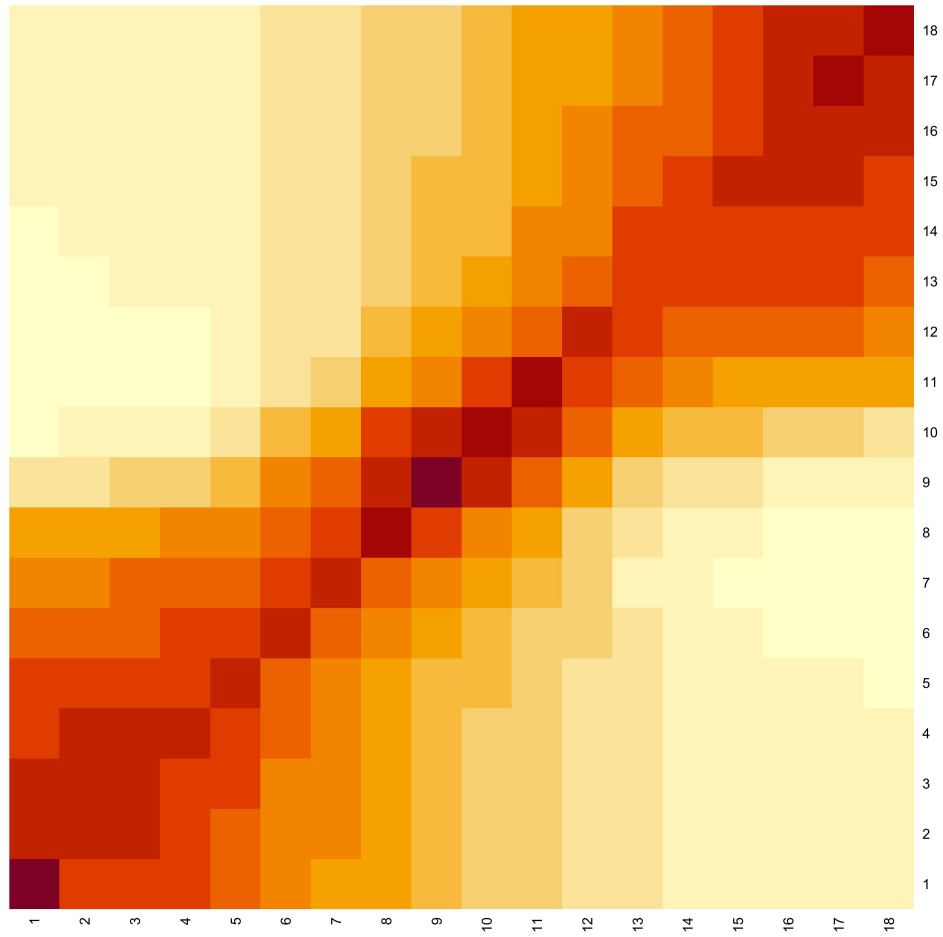


Figure 4: Population-level coancestry matrix describing the simulated stepping stones structure used in the tutorial.

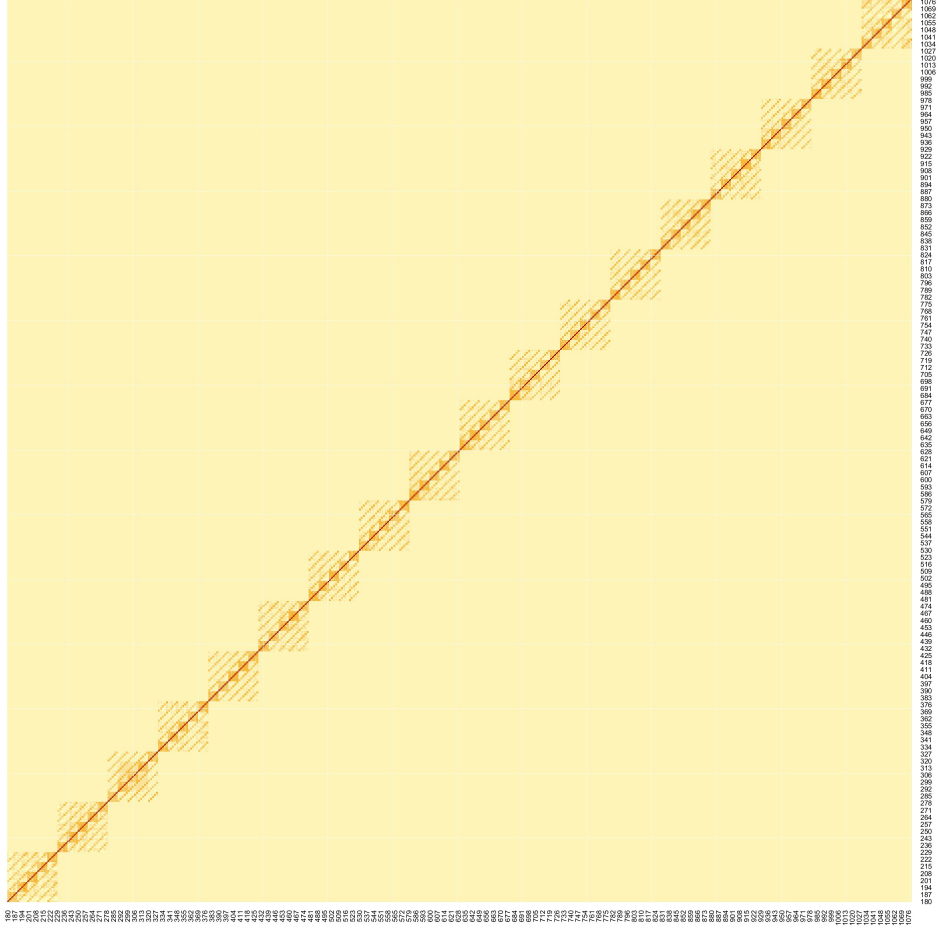


Figure 5: Individual-level coancestry matrix for the example used in our tutorial.

### 7.2.2 Individual-Level Relatedness ( $M$ )

$M$  is an  $n \times n$  matrix (where  $n$  is the number of F1 individuals) that describes genetic relationships among individuals within populations.

For common garden designs with within-population crosses,  $M$  is block-diagonal, with each block corresponding to one population.

The within-population relatedness is adjusted to account for the population-level coancestry:

$$\hat{M}_x = \text{relatedness}_x \times (1 - \hat{\Theta}_x^P)$$

where  $\hat{\Theta}_x^P$  is the diagonal element of  $\hat{\Theta}_P$  for population  $x$ .

This adjustment is critical because it ensures that  $M$  captures relatedness beyond what is already explained by population membership.

$M$  is estimated from either the F1 genetic data (using the same allele-sharing method) or the pedigree information (using standard kinship coefficients).

You can see the structure of  $M$  used in the example dataset of our tutorial in Figure 5.

## 7.3 Common Questions

### 7.3.1 Can I use the same genetic data to calculate both $M$ and $\Theta_P$ ?

Ideally, no.  $\Theta_P$  should describe the neutral structure of the source populations in the wild, while  $M$  describes relatedness among F1 individuals in the common garden. Measuring both through F1 will not consider that you have added one new generation of drift from the breeding process.

### 7.3.2 Is a common garden necessary?

Yes, we strongly recommend using a common garden or another strategy to standardize environmental effects. Like  $Q_{ST}$ - $F_{ST}$  methods, MONET results without common gardens can be confounded by phenotypic plasticity (or environmental effects) and genotype-by-environment interactions. Without environmental standardization, you cannot reliably distinguish genetic differences from environmental effects or plasticity.

## 8 Getting Help

If you encounter issues not covered in this vignette:

- **Package documentation:** `help(package = "MONET")`
- **Function help:** `?monet` or `?calculate_coancestries`
- **GitHub repository:** <https://github.com/isadoo/MONET>
- **Email contact:** [jeronimoisabela@gmail.com](mailto:jeronimoisabela@gmail.com)
- **Report issues:** <https://github.com/isadoo/MONET/issues>

For issues with `brms` specifically:

- **brms documentation:** <https://paul-buerkner.github.io/brms/>
- **Stan forums:** <https://discourse.mc-stan.org/>

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